

Spatial scRNA-seq

Jaewoong Lee Semin Lee

Department of Life Sciences
Gwangju Institute of Science and Technology

`jaewoong@gist.ac.kr`

2026-09-22

Overview

- 1 Abstract
- 2 Introduction
- 3 Materials
- 4 Methods
- 5 Results
- 6 References

1. Abstract

2. Introduction

3. Materials

4. Methods

Methods I

Basic method

- 1 Scanpy (Wolf, Angerer, & Theis, 2018)

Projection method

- 1 UMAP (McInnes, Healy, & Melville, 2018; Becht et al., 2019)
- 2 tSNE (Maaten & Hinton, 2008)
- 3 PHATE (Moon et al., 2019)
- 4 trimap (Amid & Warmuth, 2019)

Clustering

- 1 Leiden (Traag, Waltman, & van Eck, 2019)
- 2 Louvain (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008)

5. Results

5. Results

5.1. Quality check

Total counts per cell I

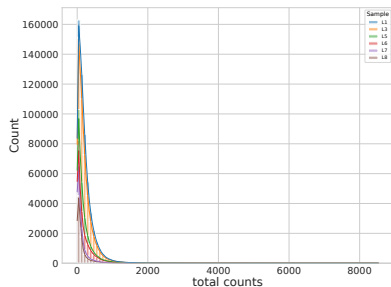


Figure: Total counts

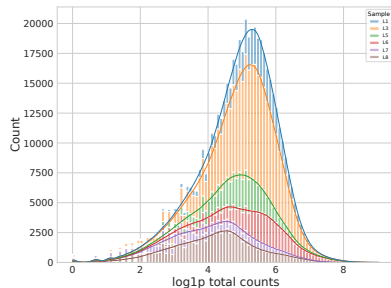


Figure: $\log_{1p}(\text{Total count})$

Total counts per cell II

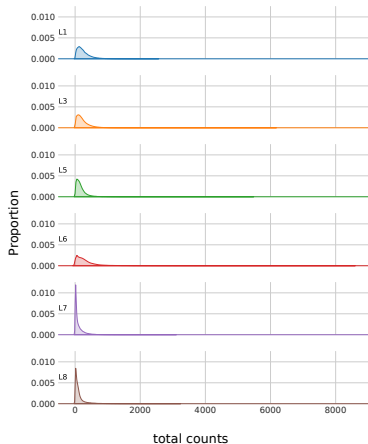


Figure: Genes by counts

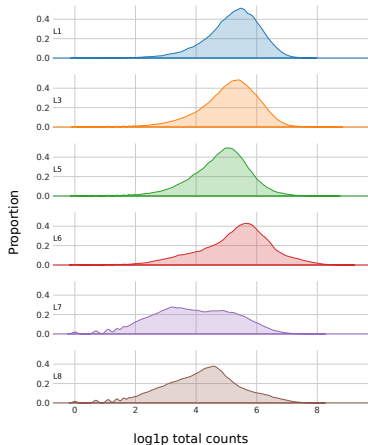


Figure: log1p(Genes by count)

Total counts per cell III

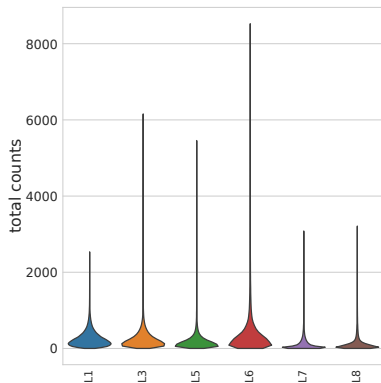


Figure: Genes by counts

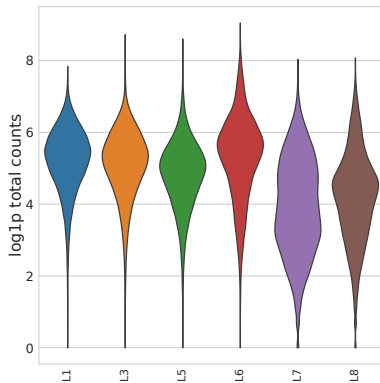


Figure: $\log_{1p}(\text{Genes by count})$

Genes by counts I

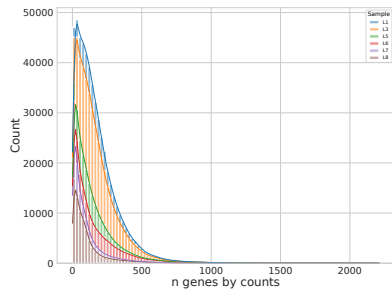


Figure: Genes by counts

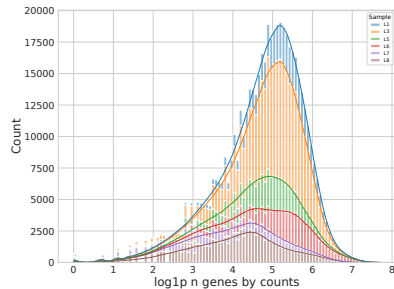


Figure: log1p(Genes by count)

Genes by counts II

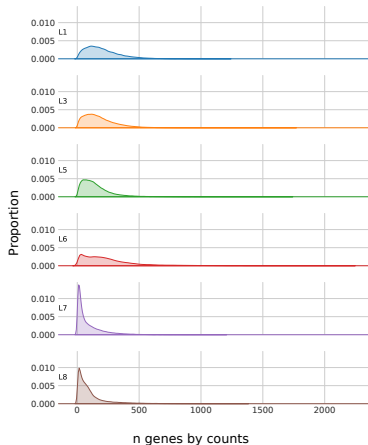


Figure: Genes by counts

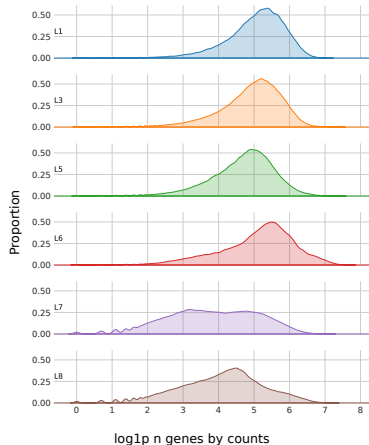


Figure: $\log_{1p}$ (Genes by count)

Genes by counts III

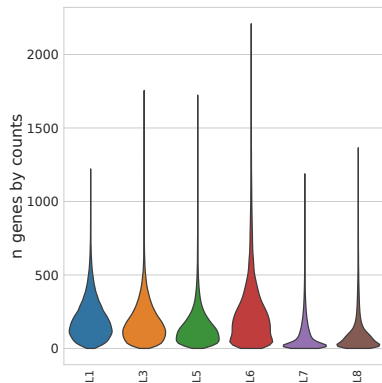


Figure: Genes by counts

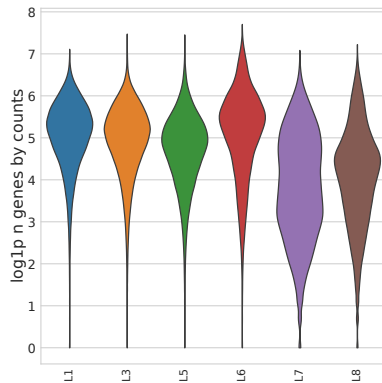


Figure: log1p(Genes by count)

5. Results

5.2. Highly variable genes

Highly variable genes with normalized dispersions I

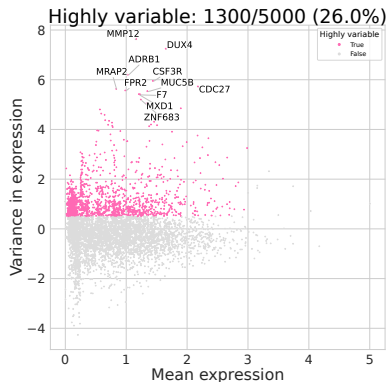


Figure: Volcano plot

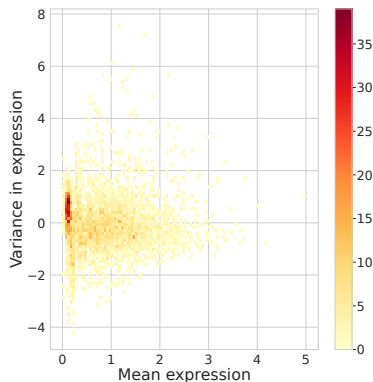


Figure: Distribution

Highly variable genes with normalized dispersions II

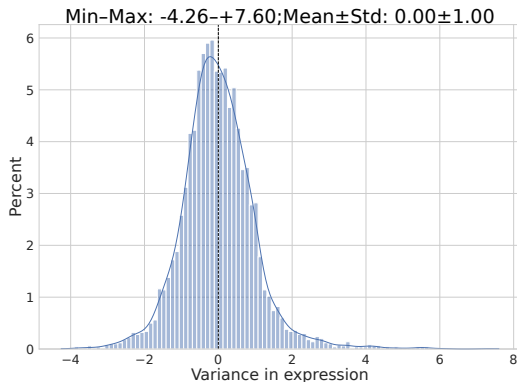


Figure: Histogram

Highly variable genes with dispersions I

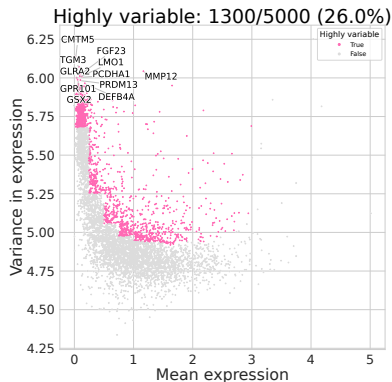


Figure: Volcano plot

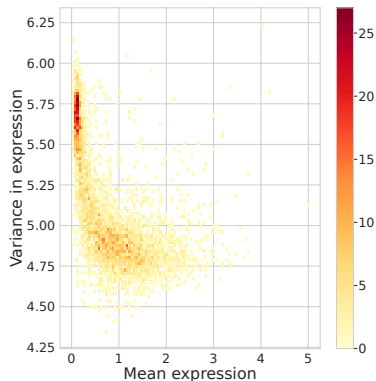


Figure: Distribution

Highly variable genes with dispersions II

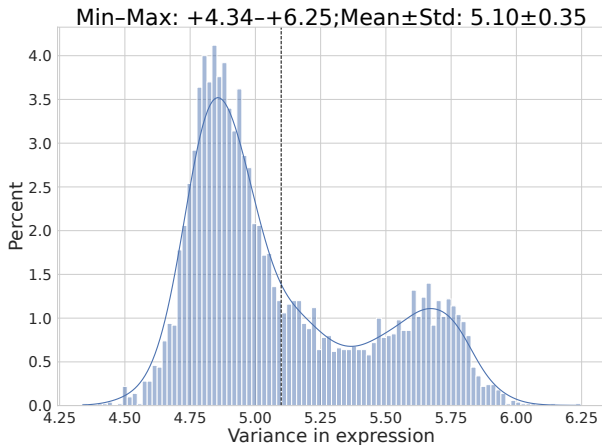


Figure: Histogram

5. Results

5.3. PCA

Explained variance

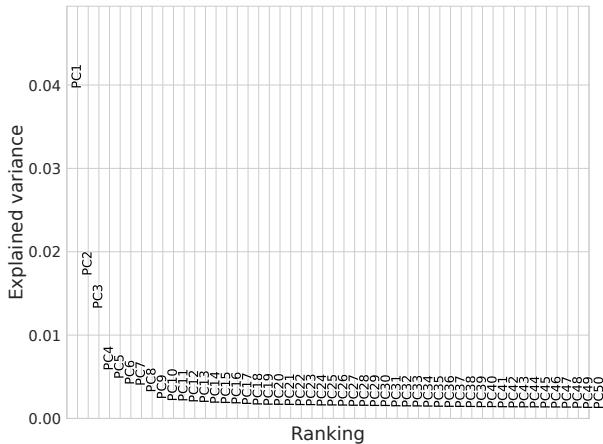


Figure: Explained variances

5. Results

5.4. Clustering

Clustering with UMAP

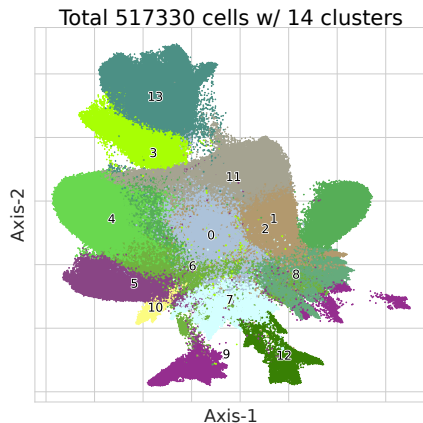


Figure: Clustering with UMAP

Cells per cluster

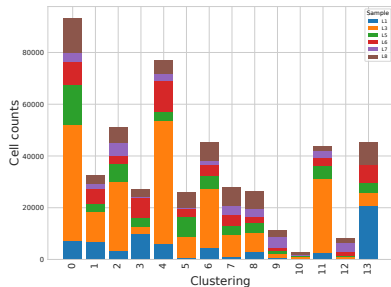


Figure: Absolute count

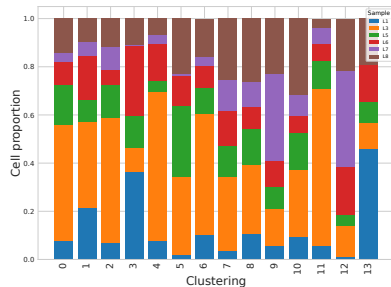


Figure: Relative proportion

Cells per sample

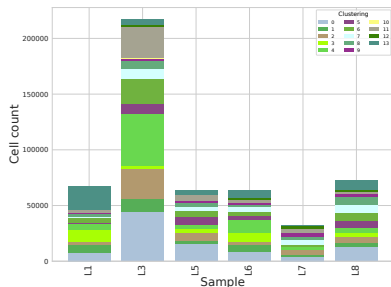


Figure: Absolute count

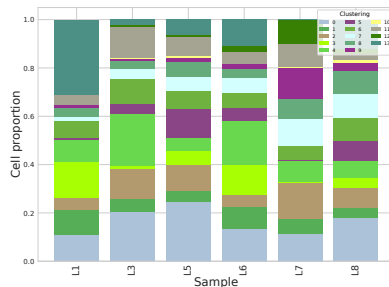


Figure: Relative proportion

5. Results

5.5. Marker genes

Cell type scoring

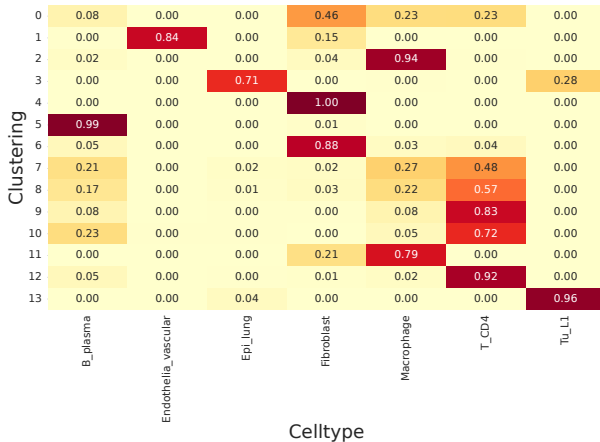


Figure: Cell type scoring

Cell types per cluster

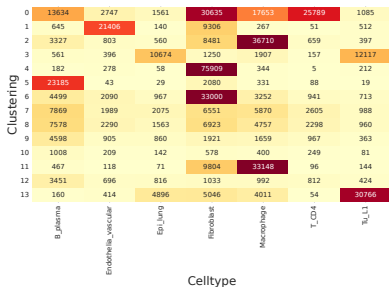


Figure: Absolute count

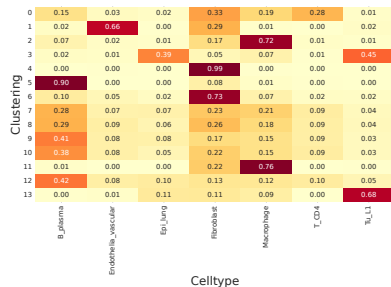


Figure: Relative proportion

5. Results

5.6. Xenimum coordiantes

6. References

- Amid, E., & Warmuth, M. K. (2019). Trimap: Large-scale dimensionality reduction using triplets. *arXiv preprint arXiv:1910.00204*.
- Becht, E., McInnes, L., Healy, J., Dutertre, C.-A., Kwok, I. W., Ng, L. G., ... Newell, E. W. (2019). Dimensionality reduction for visualizing single-cell data using umap. *Nature biotechnology*, 37(1), 38–44.
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment*, 2008(10), P10008.
- Maaten, L. v. d., & Hinton, G. (2008). Visualizing data using t-sne. *Journal of machine learning research*, 9(Nov), 2579–2605.
- McInnes, L., Healy, J., & Melville, J. (2018). Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.

- Moon, K. R., Van Dijk, D., Wang, Z., Gigante, S., Burkhardt, D. B., Chen, W. S., ... others (2019). Visualizing structure and transitions in high-dimensional biological data. *Nature biotechnology*, 37(12), 1482–1492.
- Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1), 5233.
- Wolf, F. A., Angerer, P., & Theis, F. J. (2018). Scanpy: large-scale single-cell gene expression data analysis. *Genome biology*, 19(1), 15.